Fractions: Concept & Equivalence

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. Numerator = 1 (top), denominator = 4 (bottom)

2. B — 2 (half of 4)

3. 1/2

4. 4/20

5. B — 4/10 (same denominator, so compare the numerators; 4 > 3)

6. Numerator = 3 (top), denominator = 6 (bottom)

7. 5

8. C — 5/10 (5 parts are shaded, so 5 of the 10 parts are not)

9. 3/15

10. 2/8 (same denominator, so compare the numerators)

11. B — 3 (the denominator is the bottom number)

12. 4

13. 2/10

14. A, B and D (2/6, 3/9 and 5/15 are all equivalent to 1/3; 4/9 and 2/9 are not)

15. 7/8 (same denominator, so compare the numerators)

Standard

16. 2/10 = 1/5

17. B — 24 (30 ÷ 5 = 6, 6 × 4 = 24)

18. 1/3

19. 3 2/4 (= 3 1/2)

20. C — 7/8

21. 1/8

22. 4

23. B — 1/2 (3 and 6 share a common factor of 3)

24. 2 1/2

25. 4/5

26. B — 1/3 (3/6 − 1/6 = 2/6 = 1/3)

27. 25

28. 1/2

29. B and D (5/8 and 7/10 cannot be simplified further; 3/9 = 1/3, 4/8 = 1/2, 6/9 = 2/3)

30. 7/8

Standard / Hard

31. Given away: 4; remaining: 8; each friend gets 4

32. A — 3 1/4 (9/4 = 2 1/4; + 1 = 3 1/4)

33. 1/4 of 20 = 5 eaten; left after eating = 15; after 3 more = 18

34. 3/6 = 1/2

35. A — 1/3 is bigger (1/3 = 2/6, and 2/6 > 1/6)

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36. Given away: 4; remaining: 12; each friend gets 4

37. A and B (10/12 and 15/18 both simplify to 5/6; 5/7 and 4/5 do not)

38. 11/4 = 2 3/4; + 1 = 3 3/4

39. B — 18 (1/3 of 24 = 8 eaten, leaving 16; +2 more = 18)

40. 4/5 (already in its simplest form)

41. 1/2 = 4/8. Since 4 > 3, 1/2 is bigger than 3/8

42. B — 6 (2/5 of 20 = 8 given away, remaining 12, shared between 2 friends = 6 each)

43. 1/5 of 30 = 6 eaten; left after eating = 24; after 4 more = 28

44. A and B (6/8 and 9/12 both simplify to 3/4; 2/3 and 4/5 do not)

45. 1/4 = 2/8. Since 2 > 1, 1/4 is bigger than 1/8

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Homework Worksheet — Answers

Easy / Standard

1. Numerator = 4 (top), denominator = 5 (bottom)

2. B — 4/6 (same denominator, so compare the numerators)

3. 1/2

4. 7

5. B — 4/20 (multiply numerator and denominator by 4)

6. Numerator = 1 (top), denominator = 3 (bottom)

7. 3/6 (same denominator, so compare the numerators)

8. C — 3/4 (1/4 is shaded, so 3/4 is not)

9. 5

10. 2/4

11. A — 1 (the numerator is the top number)

12. 5/8 (same denominator, so compare the numerators)

13. 2/5

14. A and B (2/10 and 3/15 both simplify to 1/5; 4/25 and 5/20 do not)

15. 3/15

Standard

16. 2 × 3 = 6

17. B — 1/2 (5 and 10 share a common factor of 5)

18. 3 3/4

19. 4/5

20. A — 1/5

21. 4 × 5 = 20

22. 1/2

23. A — 3 1/3 (10 ÷ 3 = 3 remainder 1)

24. 9/10

25. 2/10 = 1/5

26. C — 9 (24 ÷ 8 = 3, 3 × 3 = 9)

27. 1/2

28. 2 1/2

29. A and C (5/10 and 4/8 both simplify to 1/2; 3/9 = 1/3 and 2/6 = 1/3)

30. 9/10

Standard / Advanced

31. 9/4 = 2 1/4; + 1 = 3 1/4

32. 5/10 = 1/2

33. C — 12 (1/3 of 18 = 6; doubled = 12)

34. 3/6 = 1/2; ×2 top and bottom = 6/12

35. 2/3 = 6/9. Since 6 > 5, 2/3 is larger than 5/9

36. Under 5 min: 1/4 × 24 = 6; remaining = 18; half of those, 9, finish in the next minute

37. C — 10 litres (1/5 × 10 = 2L poured out, leaving 8L; 8L + 2L = 10L)

38. Spent: £10 + £5 = £15; left = £40 − £15 = £25

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39. Monday given: 1/4 × 24 = 6; left after Monday = 18; Tuesday given = half of that = 9; left after Tuesday = 9
40. A, B and D (3/4 of 16 = 12 eaten, leaving 4; 4 ÷ 2 = 2 pieces each; 4/16 = 1/4 remains)
41. Poured out: 1/3 × 12 = 4L; remaining = 8L; after topping up = 11L
42. C — £21 (book = £6, magazine = £3, spent = £9; £30 − £9 = £21)
43. Monday given: 1/3 × 18 = 6; left after Monday = 12; Tuesday given = half of that = 6; left after Tuesday = 6
44. Under 5 min: 1/4 × 16 = 4; remaining = 12; half of those = 6
45. A and B (2/5 of 20 = 8 eaten, leaving 12; 12 ÷ 4 = 3 pieces each; C is false, and 12/20 = 3/5 remains, not 1/2, so D is false)

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